

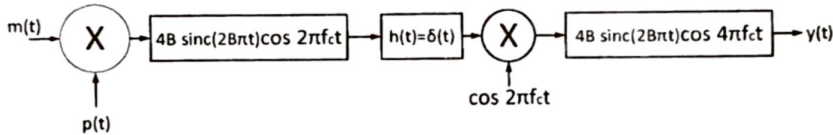
Birla Institute of Technology and Science, Pilani
ECE/EEE F311 Communication Systems
First Semester 2025-2026 (06-10-2025)
Mid-Semester Test (Closed Book)

Maximum Marks: 75

Duration: 90 minutes

- There are four questions.
 - A formula sheet consisting of FT, FS, and trigonometric identities is provided. Please do not write anything on the formula sheet.
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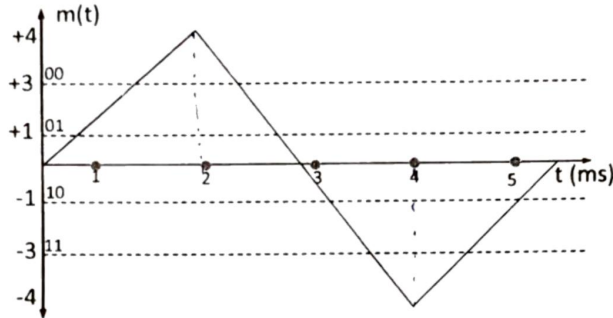
- Q1. (a) A signal $x(t) = \sin(100\pi t) + \sin(200\pi t)$ is transmitted over a distortionless channel $h(t)$ to get output signal $y(t) = \frac{1}{2} [\sin(100\pi t - 100\pi) + \sin(200\pi t + \phi)]$. Find $h(t)$ and ϕ . [5 Marks]
- (b) For the block diagram as shown in the figure below, find the signal $y(t)$, where $p(t) = \frac{4}{\pi} (\cos 2\pi f_c t - \frac{1}{3} \cos 3(2\pi f_c t) + \frac{1}{5} \cos 5(2\pi f_c t) - \dots)$. Assume $f_c \gg B$, where B is the bandwidth of the message signal. [5 Marks]



- (c) Find the power efficiency of LSSB signal $x(t) = 4 \cos 2\pi f_c t + m(t) \cos 2\pi f_c t + m_h(t) \sin 2\pi f_c t$, where the average power of the message signal $m(t)$ is 8 Watt. [5 Marks]
- (d) There are two radio stations operating: Station A at 540 kHz and Station B at 1340 kHz. Both stations transmit audio signals having a bandwidth of 4 kHz using the DSB-SC modulation scheme, and each station occupies a total guard band of 2 kHz. Draw the block diagram of a superheterodyne receiver designed to receive the signal from Station A, where Station B acts as the image station corresponding to Station A. Also, specify the center frequency and bandwidth of each block used in the receiver. [5 Marks]
- Q2. A message signal $m(t) = 2 \cos(200\pi t)$ modulates a carrier $\cos(2\pi f_c t)$ resulting the modulated signal $x(t)$. Find the following:
- (a) If the message signal amplitude modulates the carrier, find the minimum power required for transmission of the AM signal $x(t)$ for distortionless envelope detection at the receiver. [6 Marks]
- (b) If the message signal frequency modulates the carrier, find the bandwidth of the FM signal $x(t)$ using Carson rule with $k_f = 100$ Hz/Volt. [6 Marks]

- (c) For the FM signal in (b), find the minimum value of carrier frequency f_c assuming $J_n(\beta) = 0$, if $n > \beta + 1$. [8 Marks]

Q3. A periodic message $m(t)$ with amplitude ± 4 volts and 5 sample points are shown in figure below. Note that time index is in millisecond (ms). Dashed lines are quantize values with bit pattern specified. The peak values $m(t) = 4$ volts and $m(t) = -4$ volt are attained at $t = 2$ ms and $t = 4$ ms, respectively.



- (a) If the analog signal is encoded into a bit stream using uniform PCM, determine the SQNR in dB. Write down encoded PCM bit stream for the duration of 5 ms. [5 Marks]
- (b) If the analog data is encoded into a bit stream using DM, determine the ^{min.} step size required to avoid slope overload distortion, assuming the data rate generated using DM becomes double of the PCM. [5 Marks]
- (c) Determine the minimum channel bandwidth required for transmitting DM data using 64-ary PAM modulation. [7 Marks]

Q4. A pulse $p(t)$ is given below.

$$p(t) = (\text{sinc}(\pi R_s t))^2 \left[\frac{\cos(\pi \alpha_1 R_s t)}{1 - (2\alpha_1 R_s t)^2} \right] \left[\frac{\cos(\pi \alpha_2 R_s t)}{1 - (2\alpha_2 R_s t)^2} \right] \quad (1)$$

where R_s denotes the symbol rate.

- (a) Using $\alpha_1 = \alpha_2 = 0$, plot $P(f) + P(f + R_s)$, and demonstrate that the pulse $p(t)$ satisfies the Nyquist criterion for zero ISI using frequency domain criteria. Also, determine the effective roll-off factor α of $p(t)$. [8 Marks]
- (b) Assume that the pulse $p(t)$ (from equation (1)) is used for transmission over a channel with a bandwidth of 2400 Hz. The system employs 16-ary PAM modulation and is required to achieve a bit rate of $R_b = 6400$ bit/s. Given $\alpha_2 = \frac{1}{2}$, find the value of α_1 such that pulse satisfies the Nyquist criterion for zero ISI, occupies the given channel bandwidth exactly, and meets the specified bit rate requirement. [10 Marks]